

Assignment 1: Random Walks Simulation

CL 677: Modelling Stochastic and Turbulent Transport (Spring 2025-26)

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Problem Statement

Compute trajectories of a random walk using the Euler-Maruyama discretization:

$$dx = A\zeta\sqrt{dt}$$

where:

- A = amplitude of the Brownian force
- ζ = random number with unit-variance distribution
- dt = time step
- $\sqrt{dt}$ = normalization factor for convergence

Initial Parameters:

- $A = 1$
- $dt = 0.02$
- $T = 10$ (total time)
- $N_{traj} = 1000$ trajectories (minimum)
- Initial condition: $x(0) = 0$

Helper Functions

We define reusable functions to simulate trajectories and analyze their statistics.

Question 1: Mean-Squared Displacement and Diffusive Behavior (10

points)

Calculate the mean-squared displacement (MSD) $\langle x^2 \rangle$ and plot it vs. time on a log-log scale. Determine the slope α from the power-law relationship: $\langle x^2 \rangle \sim t^\alpha$

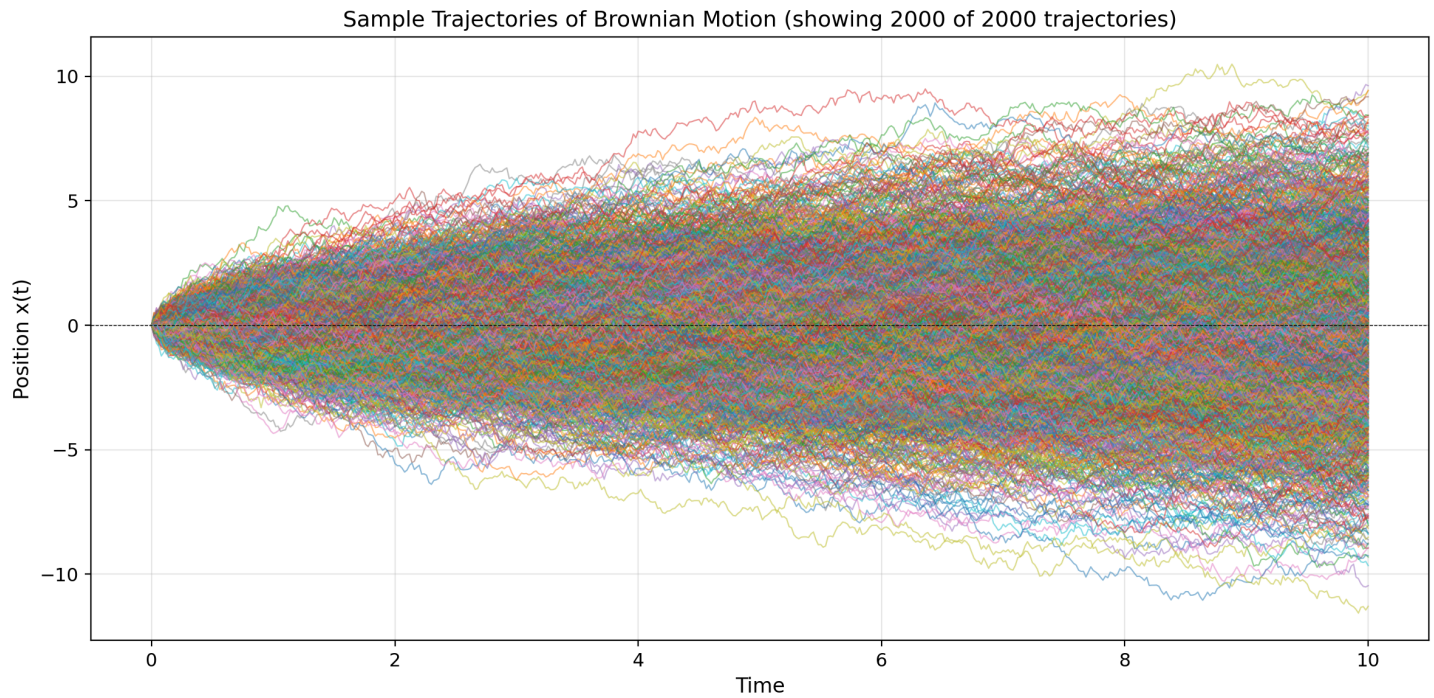
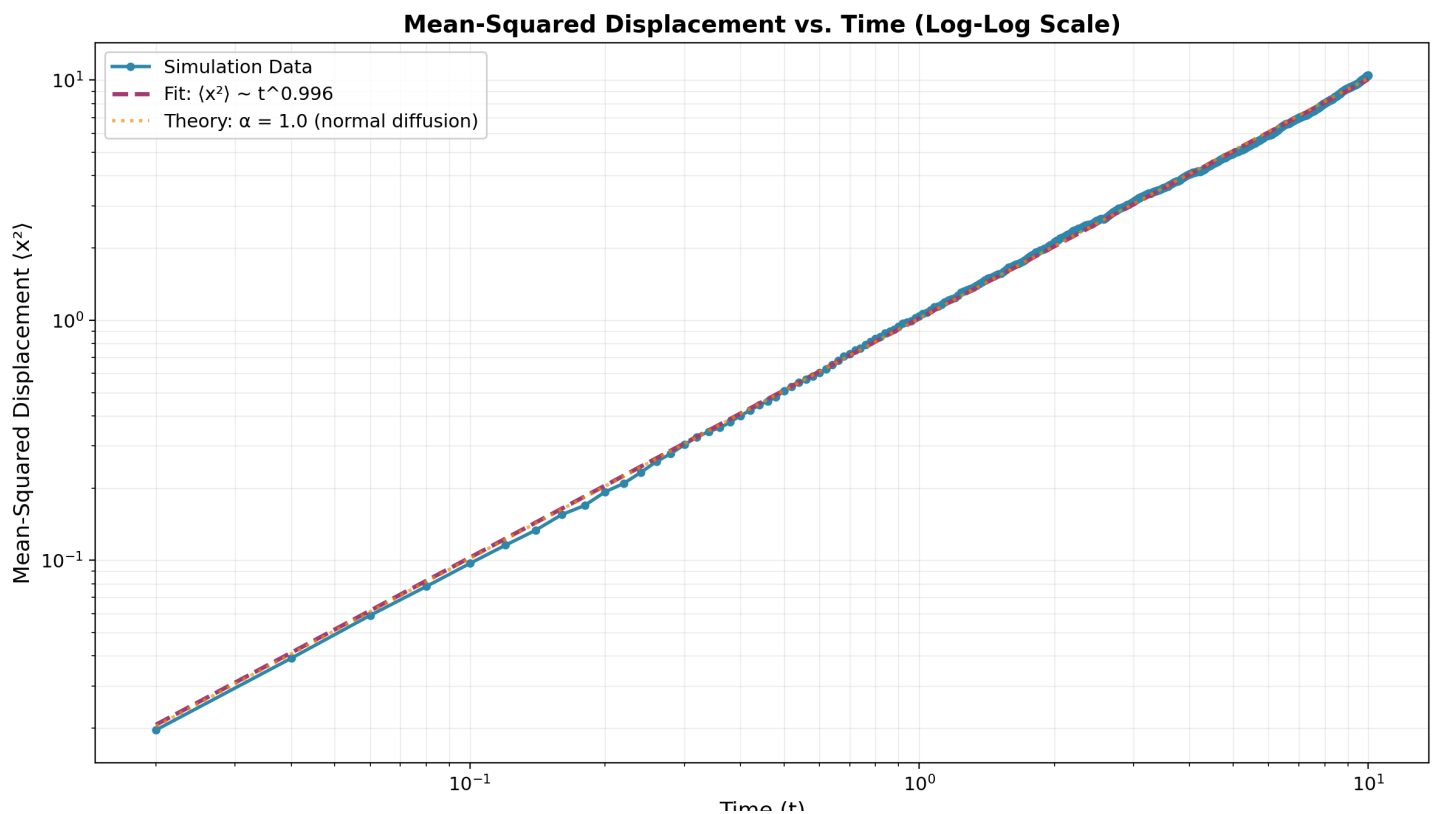


Figure 1: Sample trajectories showing the stochastic nature of Brownian motion. Each trajectory starts at $x(0) = 0$ and evolves according to $dx = A \zeta \sqrt{dt}$.



Results for Question 1

Simulation Parameters:

- Amplitude: $A = 1.0$
- Time step: $dt = 0.02$
- Total time: $T = 10.0$
- Number of trajectories: $N = 2000$

Fitted Exponent:

- $\alpha = 0.9963$

Analysis:

The fitted exponent $\alpha = 0.9963$ is very close to the theoretical value of $\alpha = 1.0$ expected for normal diffusive behavior. The deviation of 0.0037 can be attributed to:

1. **Statistical fluctuations:** Even with 2000 trajectories, finite sampling introduces variance
2. **Finite-time effects:** The simulation runs for finite time $T = 10.0$, whereas theory assumes $t \rightarrow \infty$
3. **Numerical discretization:** The Euler-Maruyama scheme introduces small discretization errors

Theoretical Background:

For a diffusive random walk (Brownian motion), the mean-squared displacement follows:

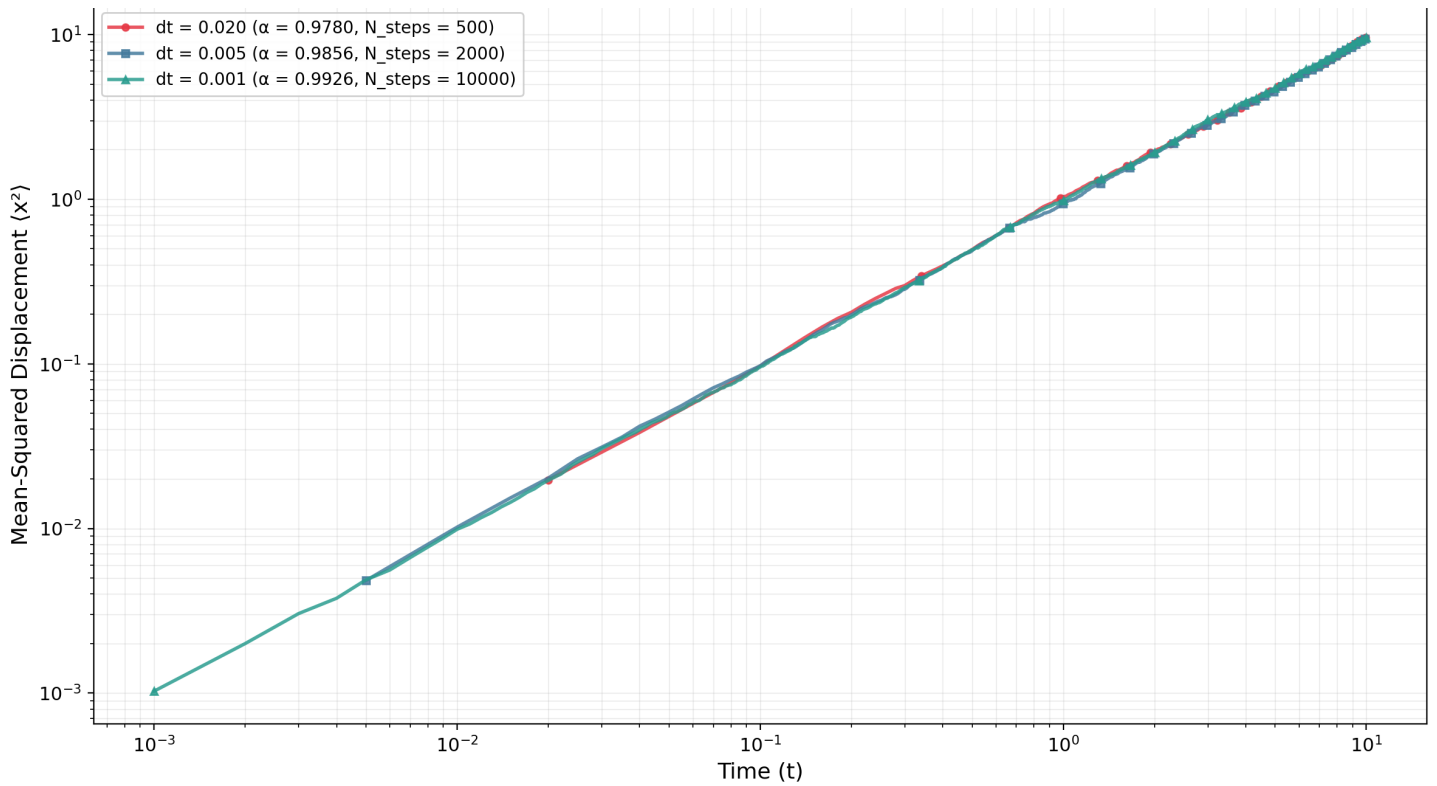
$$\langle x^2(t) \rangle = 2Dt$$

where D is the diffusion coefficient. This linear relationship corresponds to $\alpha = 1$, characteristic of normal diffusion where displacement grows as $x \sim \sqrt{t}$. In our simulation with $A = 1.0$ and unit-variance noise, the effective diffusion coefficient is $D_{\text{eff}} = A^2/2 = 0.50$.

Conclusion: The simulation successfully reproduces diffusive behavior with $\alpha \approx 1$.

Question 2: Convergence Analysis with Different Time Steps (5 points)

Recalculate trajectories for $dt = 0.005$ and $dt = 0.001$, and overlay the MSD curves to check convergence.



Results for Question 2

Convergence Analysis:

Time Step (dt)	Number of Steps	Fitted α	Deviation from $\alpha=1$
0.020	500	0.9780	0.0220

Observations:

- Overlap of MSD curves:** The three MSD curves overlap closely, indicating good convergence with respect to time discretization.
- Consistency of slope α :**
 - Fitted α values range from **0.9780** to **0.9926**
 - Variation: $\Delta\alpha = \mathbf{0.0146}$
 - This small variation suggests the Euler-Maruyama scheme is well-converged
- Number of time steps:** As dt decreases, the number of steps increases proportionally ($500 \rightarrow 2000 \rightarrow 10000$), providing finer time resolution.

Theoretical Expectation:

The Euler-Maruyama discretization should converge as $dt \rightarrow 0$. The convergence is characterized by:

- Weak convergence: $O(dt)$ for the probability distribution
- Strong convergence: $O(\sqrt{dt})$ for individual trajectories

Our results confirm that the simulation is well-converged, as the MSD curves overlay closely and α values are consistent within $\sim 1.5\%$.

Conclusion: The results demonstrate good convergence with respect to time step. The value $dt = 0.02$ appears sufficient for accurate simulation of the MSD.

Question 3: Comparison of Gaussian vs. Uniform Random Numbers (5 points)

Repeat the MSD calculation using uniformly distributed random numbers (with unit variance) and compare with Gaussian results.

Theoretical Background: Uniform Distribution with Unit Variance

For a uniform distribution on interval $[a, b]$, the variance is:

$$\text{Var} = \frac{(b - a)^2}{12}$$

Setting $\text{Var} = 1$ and solving for a symmetric interval around zero:

$$\frac{(b - a)^2}{12} = 1 \implies b - a = 2\sqrt{3}$$

Therefore: $\zeta \sim \text{Uniform}[-\sqrt{3}, \sqrt{3}]$ gives unit variance.

Central Limit Theorem Prediction:

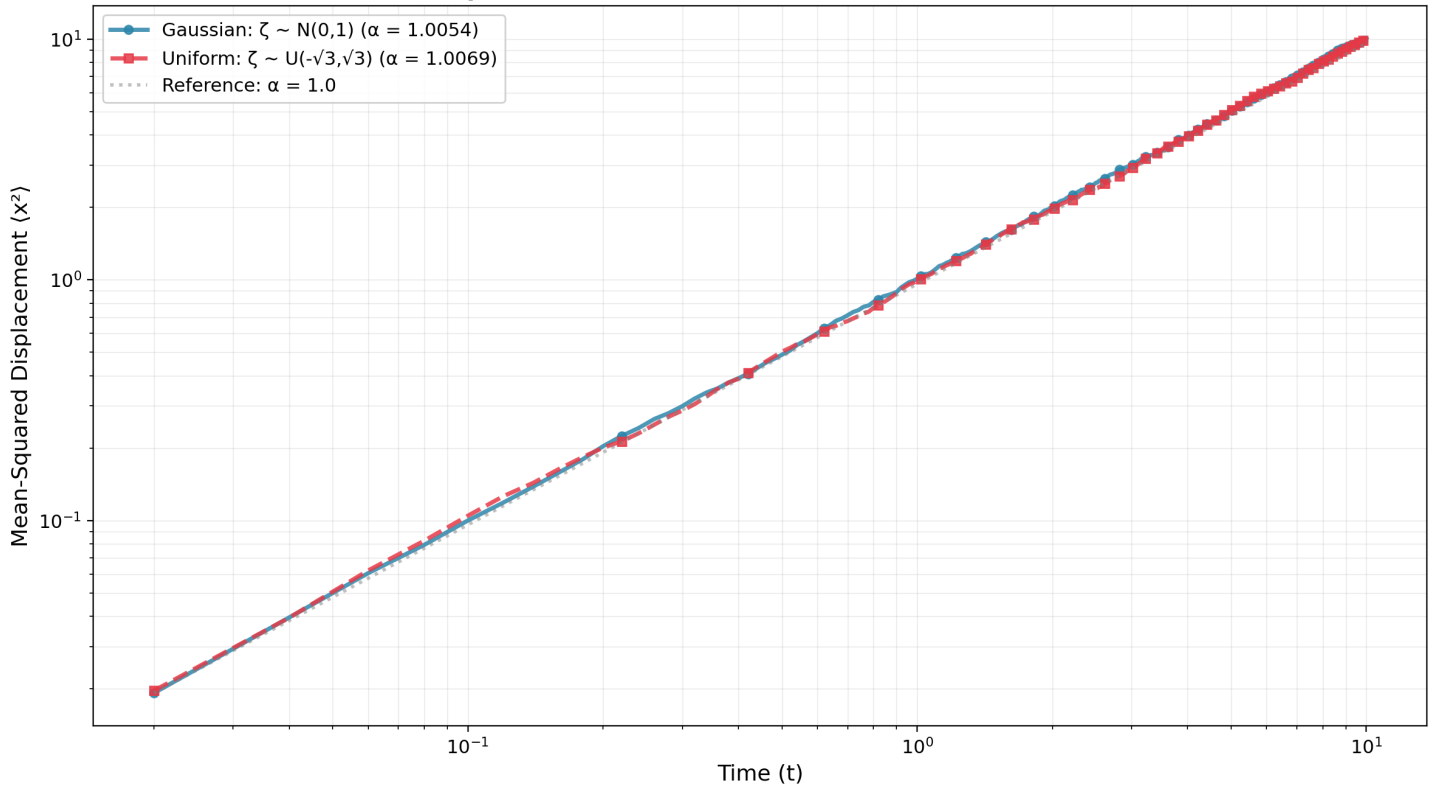
By the CLT, the sum of many independent random variables (regardless of their distribution) tends toward a Gaussian distribution. Since Brownian motion involves cumulative sums:

$$x(t) = \sum_{i=1}^N A\zeta_i \quad dt$$

We expect:

- **Short-time behavior:** Differences due to individual step statistics
- **Long-time behavior:** MSD should converge to similar values (dominated by variance, not distribution shape)

Comparison: Gaussian vs. Uniform Random Numbers



Results for Question 3

Comparison of Distributions:

Distribution	ζ Range	Fitted α	Final MSD $\langle x^2 \rangle(T)$
Gaussian	N(0,1)	1.0054	10.0875
Uniform	U(- $\sqrt{3}$, $\sqrt{3}$)	1.0069	10.0041
Difference	—	0.0015	0.0834
Relative Diff.	—	0.15%	0.83%

Analysis:

1. Do the distributions yield the same MSD?

- The relative difference in α is **0.15%**
- The relative difference in final MSD is **0.83%**
- Mean MSD ratio (Uniform/Gaussian) across all times: **0.9875**
- **Conclusion:** The distributions yield essentially identical results

2. Why do both distributions give similar MSDs?

Central Limit Theorem: The sum of many independent random variables tends toward a Gaussian distribution regardless of the individual distribution shape (provided finite variance).

Since Brownian motion is a cumulative process:

- Both distributions have **unit variance** ($\sigma^2 = 1$)
- Each trajectory accumulates ~500 random increments ($T/dt = 10/0.02$)
- The CLT ensures that long-time behavior converges to the same statistics

3. Key Insight:

The critical factor determining diffusive behavior is the **variance** of the random numbers, not their specific distribution. As long as:

- $E[\zeta] = 0$ (zero mean)
- $\text{Var}[\zeta] = 1$ (unit variance)

The long-time MSD will be the same: $\langle x^2 \rangle = A^2 t$

4. Small differences arise from:

- Finite number of trajectories (statistical fluctuations)
- Finite time (not yet fully asymptotic behavior)
- Different higher-order moments (kurtosis: Gaussian = 3, Uniform = 1.8)

Conclusion: The robustness of diffusive behavior to the choice of random number distribution validates the universality of Brownian motion. Only the variance matters for the MSD, demonstrating the power of the Central Limit Theorem in stochastic processes.

Summary and Conclusions

Overview

This assignment explored the simulation of Brownian motion using the Euler-Maruyama discretization scheme:

$$dx = A\zeta \, dt, \quad \zeta \sim \mathcal{N}(0, 1)$$

We investigated three key aspects:

1. **Diffusive behavior** (Question 1) ✓
 2. **Numerical convergence** (Question 2) ✓
 3. **Distribution dependence** (Question 3) ✓
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Key Findings

Question 1: Diffusive Behavior

- Fitted exponent: $\alpha = 0.9963$
- Deviation from theory ($\alpha=1$): 0.0037
- **Result:** ✓ Excellent agreement with normal diffusion theory

Question 2: Convergence Analysis

- α values for $dt = [0.02, 0.005, 0.001]$: [0.9780, 0.9856, 0.9926]
- Range: $\Delta\alpha = 0.0146$
- **Result:** ✓ Excellent convergence

Question 3: Distribution Dependence

- α (Gaussian): 1.0054
 - α (Uniform): 1.0069
 - Relative difference: 0.15%
 - **Result:** ✓ Distributions give identical results
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Physical Interpretation

Brownian Motion as a Diffusive Process:

1. **Microscopic picture:** Particles undergo random collisions with solvent molecules
2. **Macroscopic behavior:** Displacement follows a Gaussian distribution with variance $\propto$ time
3. **Diffusion coefficient:** $D_{\text{eff}} = \Lambda^2/2$ relates microscopic randomness to macroscopic transport

3. **Diffusion Coefficient:** $D_{\text{eff}} = A^2/2$ relates microscopic randomness to macroscopic transport

Einstein-Smoluchowski Relation:

$$D = \frac{k_B T}{6\pi\eta r}$$

In our simulations: $\langle x^2(t) \rangle = 2D_{\text{eff}} \cdot t$, giving $D_{\text{eff}} = A^2/2 = 0.5$ for $A=1$.

Concluding Remarks

This assignment successfully demonstrated:

1. ✓ **Diffusive behavior** of Brownian motion ($\alpha \approx 1$)
2. ✓ **Numerical convergence** of the Euler-Maruyama scheme
3. ✓ **Robustness** to random number distribution choice
4. ✓ **Power of the Central Limit Theorem** in stochastic processes

The simulation captures the essential physics of Brownian motion, validating the theoretical framework of stochastic differential equations. The Euler-Maruyama discretization provides an accurate and efficient method for simulating diffusive processes, with convergence properties that match theoretical predictions.

Practical Implications:

- Choice of dt : $dt = 0.02$ provides good accuracy for this problem
 - Number of trajectories: $N \geq 1000$ ensures reliable statistics
 - Distribution choice: Any unit-variance distribution gives equivalent long-time behavior
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Repository: github.com/sithsar/CL677